When Do We Plan Agreement?

Evidence from Unaccusatives and Agreement Attraction

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Attraction is not automatic

Error rates range from <2% to ~10% across four experiments with similar structures. Task design, not syntax alone, determines whether attraction surfaces.

Controller viability gates interference

Attraction surfaces only when the attractor is a viable controller candidate, paralleling response set effects in production and controllerhood effects in comprehension.

Timing reveals hidden costs

Even when errors are suppressed, preverbal times and onset times can reveal residual processing difficulty from number mismatch.

Unaccusatives reveal early planning

Onset times correlate with attraction errors only for unaccusatives, consistent with earlier verb planning exposing morphosyntactic conflict at utterance initiation.

The Puzzle

- **Agreement attraction:** verbs sometimes agree with a non-subject noun (the *attractor*) instead of the true controller (Bock & Miller, 1991).
- Existing accounts treat attraction as pervasive: **representational** (attractor corrupts the subject's number) and **retrieval-based** (wrong noun retrieved at the verb). Both predict attraction whenever mismatching nouns are structurally present.

But attraction rates in scene description vary from 3% to 25% across studies with the same syntactic structure (Nozari & Omaki, 2022; Kandel & Phillips, 2022). This five-fold difference dwarfs any within-study manipulation (typically 2–4%).

Is attraction architectural noise, or does it depend on whether the task makes the attractor a viable controller candidate?

Controller Viability

- **Response set effects:** distractors interfere only when they belong to the eligible response set (La Heij & Van den Hof, 1995; Roelofs, 2001; Lamers et al., 2010).
- Paralleled in comprehension: a noun's statistical regularity in controller positions modulates its ability to interfere (Lago et al., 2019; Bhatia & Dillon, 2022).

Our proposal: Agreement operates over a task-defined set of viable controller candidates.

Attractors interfere only when included in this set, gated by visual layout, candidate pool, and noun–verb mapping consistency.

Multiple Signatures

- Floor-level error rates could reflect genuinely absent interference or merely suppressed errors with residual processing costs. We need **multiple measures** to distinguish these.
- **Preverbal pause duration:** indexes agreement planning difficulty even when errors are avoided (Kandel & Phillips, 2022), as in '*The key to the cabinets ..... are rusty*'.
- **Utterance onset time:** captures early planning difficulty at articulation initiation. It is possible that for verbs that are planned early, we could see pre-utterance effect (Momma & Ferreira, 2019).

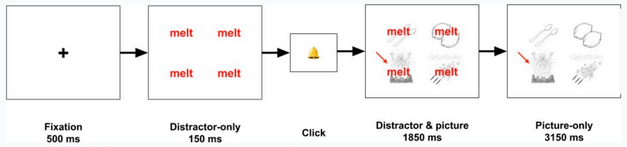
If a manipulation eliminates all signatures, attraction is gated by controller viability. If errors disappear but timing effects persist, the mechanism is engaged but errors are resolved.

Selected References: Bhatia & Dillon (2022) *JML* · Bock & Miller (1991) *Cog Psych* · Eberhard, Cutting, & Bock (2005) *JML* · Kandel & Phillips (2022) *JML* · La Heij & Van den Hof (1995) *JEP:LMC* · Lago et al. (2015) *JEP:LMC* · Lago et al. (2019) *Ling Approaches to Bilingualism* · Lamers, Roelofs, & Rabeling-Keus (2010) *Acta Psych* · Momma & Ferreira (2019) *JEP:LMC* · Roelofs (2001) *Cognition* · Wagers, Lau, & Phillips (2009) *JML*

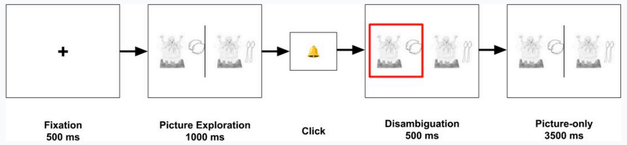
Experimental Design

	EXP 1	EXP 2	EXP 3	EXP 4
Comm. intent	No	Yes	Yes	Yes
Viable ctrl?	No	No	Yes	Yes
Attractors	Inanimate	Inanimate	Animate	Animate
Attraction	1.6%	6.0%	9.4%	11.0%

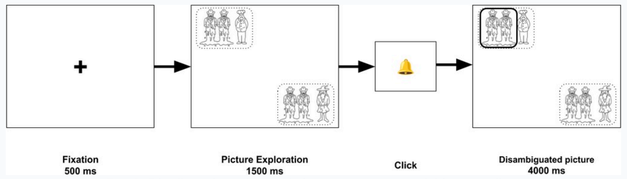
Scene description task (head number x attractor number x verb type).
Participants produce: *the **HEAD** prep the **ATTRACTOR** is/are verb-ing.*



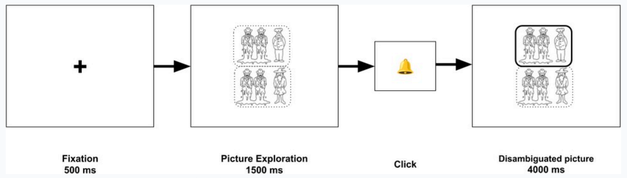
Exp 1. Centralized grid, red arrow cue. Inanimate attractors, no communicative need.



Exp 2. Side-by-side scenes, red square cue. Attractor needed to disambiguate.



Exp 3. Distributed quadrants. Animate attractors as viable controllers.

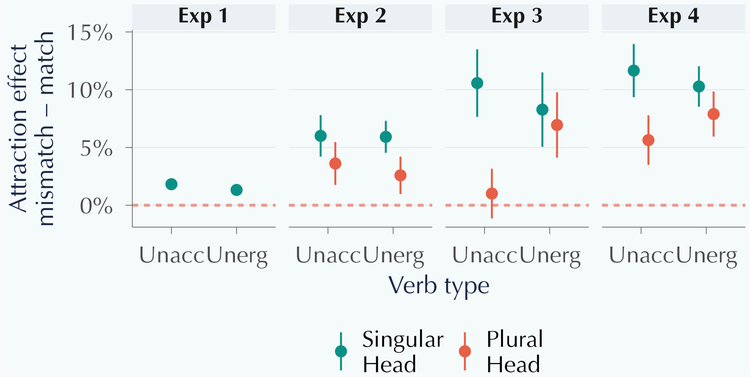


Exp 4. Centralized grid, holistic cue encompasses target + attractor.

Key Conclusions

- **Attraction is not pervasive noise.** It is a concentrated, high-probability outcome that emerges when specific conditions make attractors viable controllers.
- **Task-level gating.** The same syntactic structure can yield very attenuated attraction magnitudes (~2%) or magnitudes comparable to comprehension (~10%) across four experiments depending whether the task elevates the attractor to controller-candidate status.
- **Timing-related probing.** Preverbal times index mismatch difficulty even when errors are suppressed following Kandel & Phillips (2022). Onset times correlate with errors selectively for unaccusatives, consistent with earlier verb planning exposing morphosyntactic conflict.

Attraction Effects

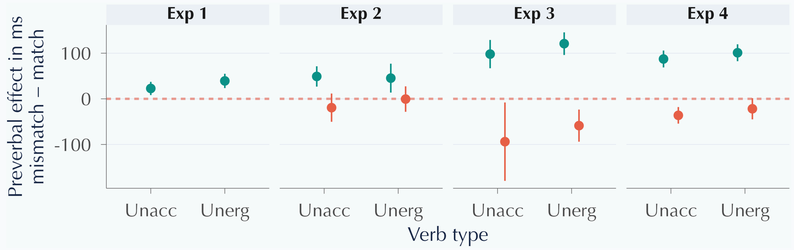


EXP 1 Near **floor**: Unacc (1.8% ± 0.5) and Unerg (1.3% ± 0.5). Attractor outside viable controller set.

EXP 2 Unacc (6.0% ± 1.8) and Unerg (5.9% ± 1.4): same verb for both scenes increases communicative need for the attractor.

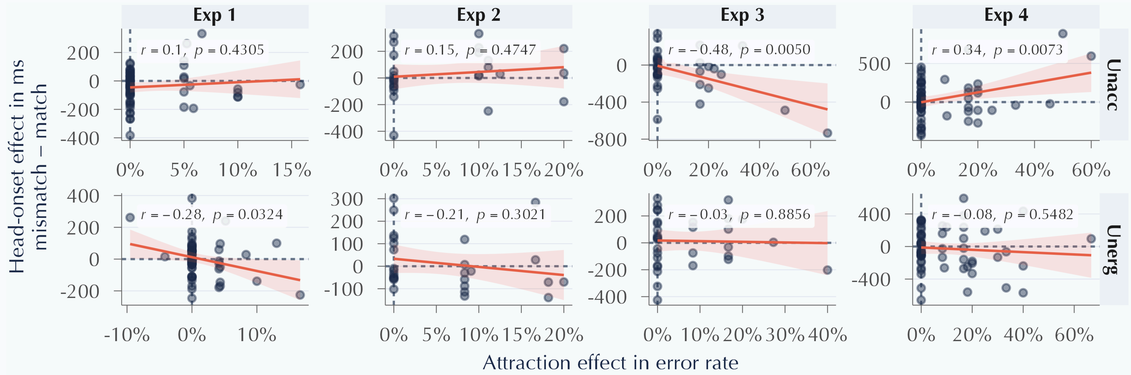
EXPS 3–4 Unacc (10.6–11.7% ± 2.3–2.9) and Unerg (8.3–10.3% ± 1.7–3.2): response set inclusion brings attraction to canonical levels.

Preverbal Time Effects



Preverbal time indexes **agreement planning difficulty**. Mismatch effects in preverbal time do not differ by verb type, consistent with late, verb-type-independent agreement morphophonology. However, it mimics the attraction effects, pointing towards non-pervasive analysis of attraction

Onset Time × Errors



CORRELATED SIGNATURES Onset time and attraction errors are **correlated**, but **only for unaccusatives**, not unergatives. Unaccusative verbs are planned earlier (Momma & Ferreira, 2019), so onset captures early morphosyntactic planning that is sensitive to number mismatch.

PREVERBAL NULL **Preverbal time does not correlate with errors** for either verb type: Unacc (r = .07, p = .50) and Unerg (r = .14, p = .16). Agreement morphophonology is planned late and independently of verb type.